

PYTHON

Module 3: Syntax & Structure

CODING

The Core Philosophy of Python Syntax

Python's syntax is built on the principle of readability. While many languages use complex symbols like semicolons (;) or curly braces ({}), Python uses English keywords and clean spacing.

Two ways to execute syntax:

1. REPL (Interactive): Type `print('Hello')` directly into the command line for immediate results.
2. Scripting: Save your logic in a `.py` file (e.g., `logic.py`) and run it using the command `python logic.py`.

Mastering Python Indentation

In Python, indentation is not just for making the code look pretty—it is a functional requirement. It tells the interpreter which lines belong to which block of code.

Crucial Rules/Errors:

- Missing Indentation: Skipping a space after a colon will result in an `IndentationError``.
- Consistency: You must use the same number of spaces throughout a single block. While 1 space is the minimum, the industry standard is 4 spaces per level.
- Example of Success:

```
if 5 > 2:
```

```
    print('Correct Indentation')
```

Variable Declaration

Unlike Java or C++, Python has no command for declaring a variable with a specific type. A variable is created the moment you first assign a value to it.

Dynamics:

- ``x = 10`` (x is now an integer)
- ``name = 'Hive'`` (name is now a string)

Python is 'dynamically typed,' meaning the same variable can even change types later if needed, though this should be used cautiously.

Documenting with Comments

Comments are essential for explaining 'why' code does something. In Python, comments start with a hash character (#).

Usage types:

- Internal Notes: ``# This section processes user login``
- Disabling Code: ``# print('temp debug')``

Python ignores everything on a line after the # sign, making it the perfect tool for documentation.

CODE

Common Syntax Pitfalls

Beginners often struggle with these three specific areas:

- Colon usage: Forgetting the `:` at the end of an `if`, `for`, or `def` line.
- Case Sensitivity: `Variable` and `variable` are two different things.
- Quote Matching: Always close your strings with the same type of quote (single or double) that you started with.

Syntax in Practice: A Summary

To write effective Python, remember: 'Flat is better than nested' and 'Readability counts.'

Practice Task:

Try nested indentation by checking if a number is positive, and if so, check if it is greater than 100. This requires two levels of consistent spacing.

CodeHive Tip: Use a modern editor like VS Code which automatically handles indentation for you!

CODE

Knowledge Check

Test your understanding of the core syntax rules discussed in this module:

Question: Is indentation in Python for readability only?

Answer: False. It is a structural requirement. Without it, the code will fail to execute.

Next Module: Delving deeper into Python Variables and Data Types.